

Collections (ArrayList, HashSet, HashMap)

Q1.

Create a Java class called Song with fields: title, artist, and duration. Store 5 Song objects in an ArrayList and print the details of each song.

Answer:

```
import java.util.ArrayList;

class Song {
    String title;
    String artist;
    int duration; // in seconds

    public Song(String title, String artist, int duration) {
        this.title = title;
        this.artist = artist;
        this.duration = duration;
    }
}

public class SongListDemo {
    public static void main(String[] args) {
        ArrayList<Song> songs = new ArrayList<>();
        songs.add(new Song("Blinding Lights", "The Weeknd", 200));
        songs.add(new Song("Kesariya", "Arijit Singh", 268));
        songs.add(new Song("Flowers", "Miley Cyrus", 200));
        songs.add(new Song("Apna Bana Le", "Arijit Singh", 252));
        songs.add(new Song("Shape of You", "Ed Sheeran", 233));

        for (Song song : songs) {
            System.out.println(song.title + " by " + song.artist + " (" + song.duration + " sec)");
        }
    }
}

/* Sample Output:
Blinding Lights by The Weeknd (200 sec)
Kesariya by Arijit Singh (268 sec)
Flowers by Miley Cyrus (200 sec)
Apna Bana Le by Arijit Singh (252 sec)
Shape of You by Ed Sheeran (233 sec)
*/
```

Q2.

Build a Java program that uses a HashSet to store unique usernames for a Zomato-style food review app. Add at least 7 usernames, including 2 duplicates, and display all unique usernames.

Answer:

```
import java.util.HashSet;

public class UniqueUsernames {
    public static void main(String[] args) {
        HashSet<String> usernames = new HashSet<>();
```

```

        usernames.add("foodie_priya");
        usernames.add("hungry_raj");
        usernames.add("tasty_tales");
        usernames.add("foodie_priya");    // duplicate
        usernames.add("spice_lover");
        usernames.add("hungry_raj");      // duplicate
        usernames.add("chef_anita");
        usernames.add("snack_attack");

        System.out.println("Total usernames added (with duplicates): 8");
        System.out.println("Unique usernames stored: " + usernames.size());
        System.out.println("Usernames:");
        for (String name : usernames) {
            System.out.println("- " + name);
        }
    }
}

/* Sample Output (order may vary since HashSet is unordered):
Total usernames added (with duplicates): 8
Unique usernames stored: 6
Usernames:
- foodie_priya
- hungry_raj
- tasty_tales
- spice_lover
- chef_anita
- snack_attack
*/

```

Q3.

Simulate a Flipkart shopping cart by creating an ArrayList of Product objects (with name and price). Write a method to calculate and print the total cart value.

Answer:

```

import java.util.ArrayList;

class Product {
    String name;
    double price;

    public Product(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

public class ShoppingCart {

    public static double calculateCartTotal(ArrayList<Product> cart) {
        double total = 0;
        for (Product p : cart) {
            total += p.price;
        }
        return total;
    }

    public static void main(String[] args) {
        ArrayList<Product> cart = new ArrayList<>();
        cart.add(new Product("Wireless Mouse", 599));
    }
}

```

```

        cart.add(new Product("Laptop Bag", 899));
        cart.add(new Product("USB-C Cable", 249));
        cart.add(new Product("Power Bank", 1299));

        System.out.println("Cart Items:");
        for (Product p : cart) {
            System.out.println("- " + p.name + ": Rs. " + p.price);
        }

        System.out.println("Total Cart Value: Rs. " + calculateCartTotal(cart));
    }
}

/* Sample Output:
Cart Items:
- Wireless Mouse: Rs. 599.0
- Laptop Bag: Rs. 899.0
- USB-C Cable: Rs. 249.0
- Power Bank: Rs. 1299.0
Total Cart Value: Rs. 3046.0
*/

```

Q4.

Use a HashMap to map Instagram usernames to their follower counts. Insert at least 5 entries, then write code to display the username(s) with the highest follower count. Hint: Loop through the entry set to find the max value.

Answer:

```

import java.util.HashMap;
import java.util.Map;

public class TopFollower {
    public static void main(String[] args) {
        HashMap<String, Integer> users = new HashMap<>();
        users.put("foodie_priya", 12000);
        users.put("travel_with_raj", 45000);
        users.put("fit_arjun", 8000);
        users.put("style_by_neha", 67000);
        users.put("gamerzone", 21000);

        String topUser = null;
        int maxFollowers = -1;

        for (Map.Entry<String, Integer> entry : users.entrySet()) {
            if (entry.getValue() > maxFollowers) {
                maxFollowers = entry.getValue();
                topUser = entry.getKey();
            }
        }

        System.out.println("User with the highest followers: " + topUser + " (" + maxFollowers + " followers)");
    }
}

/* Sample Output:
User with the highest followers: style_by_neha (67000 followers)
*/

```

Q5.

Explain in your own words (2-3 sentences) how using collections like ArrayList or HashSet in Java is similar to how apps like Spotify or BookMyShow manage lists of songs, movies, or users in memory before saving to a database.

Answer:

When an app like Spotify loads your playlist or BookMyShow loads today's list of showtimes, it usually pulls that data from the database once and keeps it in memory as a collection - an ArrayList when order and duplicates matter (like the sequence of songs in a playlist), or a HashSet when only uniqueness matters (like a set of seat numbers already booked). Working with this in-memory collection is much faster than hitting the database on every single action, so the app only writes back to the database periodically or when the user makes an actual change - collections act as a temporary, fast working copy of the data the app is actively using.